



GS FOUNDATION 2024-25

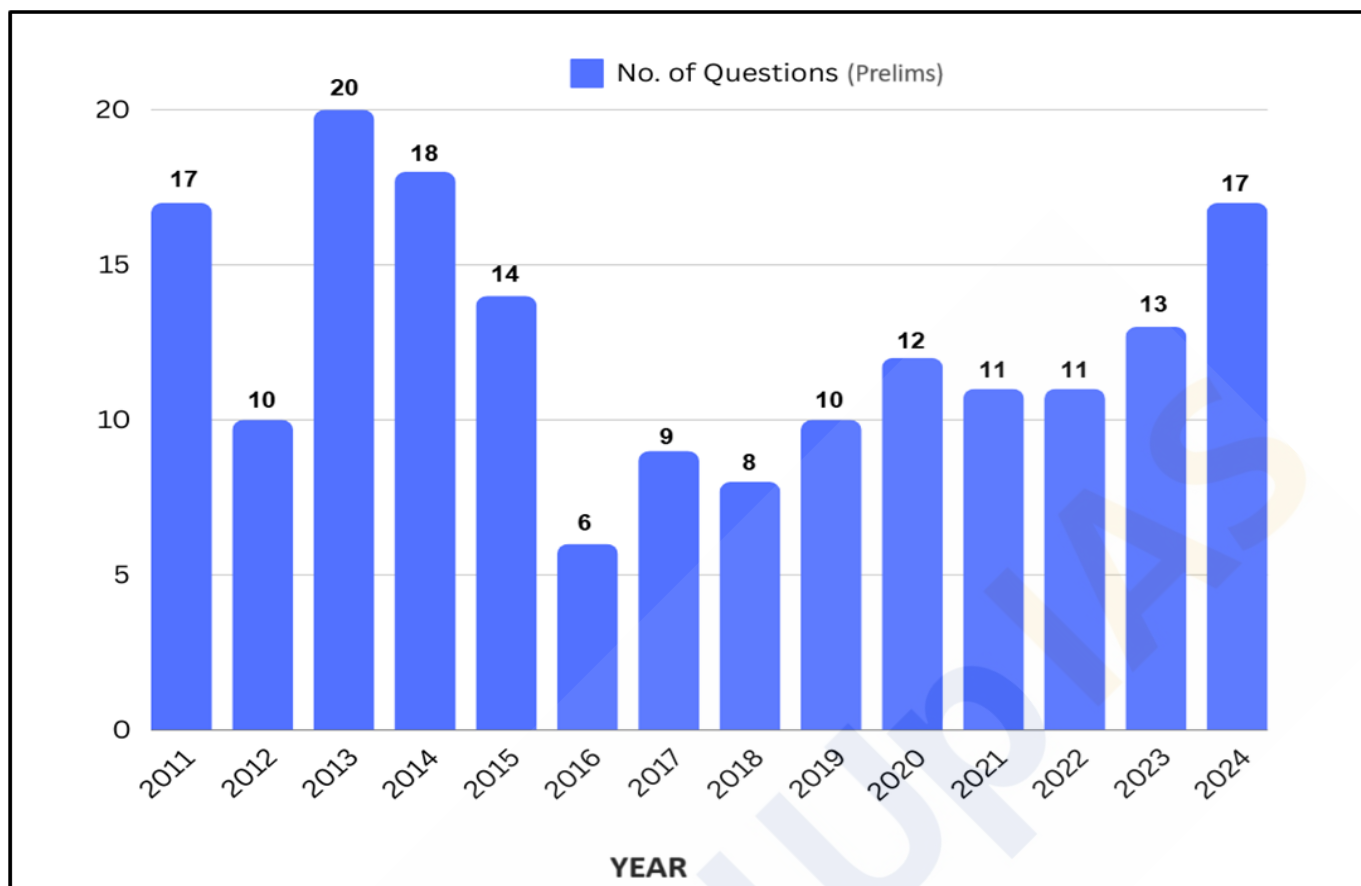
GEOGRAPHY - 18

Geomorphic Processes-II

TABLE OF CONTENTS

Landslides	3
<i>Types of Landslides</i>	<i>3</i>
<i>Factors Affecting Landslides</i>	<i>4</i>
<i>Landslide Prevention and Mitigation</i>	<i>5</i>
<i>Debris Avalanche and Landslide in India</i>	<i>5</i>
Erosion	7
<i>Soil Erosion</i>	<i>7</i>
Types of Soil Erosion	7
<i>Water Erosion</i>	<i>7</i>
<i>Wind Erosion</i>	<i>8</i>
<i>Glacial Erosion</i>	<i>9</i>
<i>Factors affecting Soil Erosion</i>	<i>9</i>
<i>Positive Impacts of Soil Erosion</i>	<i>10</i>
<i>Negative Impacts of Soil Erosion</i>	<i>10</i>
<i>Methods of Controlling Soil Erosion</i>	<i>12</i>
Deposition	12

Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

Indian Geography

Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10

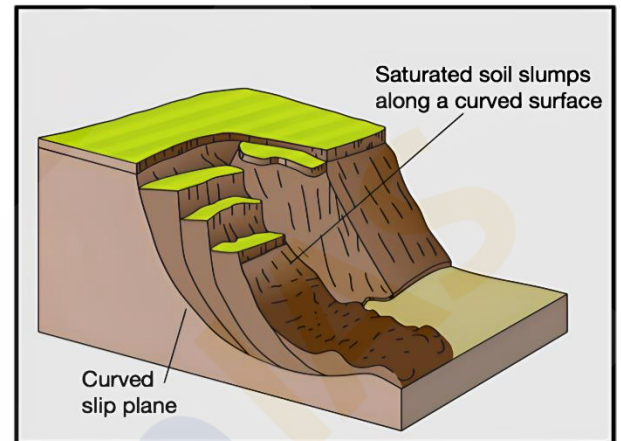
Landslides:

Landslides refer to the **downward movement of rock, soil, and other debris** on a slope due to gravity. They can be triggered by natural events such as **heavy rainfall, earthquakes, volcanic activity**, or human activities like **deforestation** and **construction**. They can occur suddenly and without warning, causing significant damage to property and loss of life. The type of landslide depends on several factors, including the type of material involved, the slope angle, the presence of water, and the triggering mechanism.

Types of Landslides:

Slump:

A slump is a type of landslide characterized by the **downward and outward movement (backward rotation) of a mass of rock or unconsolidated material** along a curved surface. This rotational movement typically results in a **concave-upward slip surface**, and the slumped material often forms a series of stepped terraces, or scarps.



Slumps are commonly triggered by the following factors:

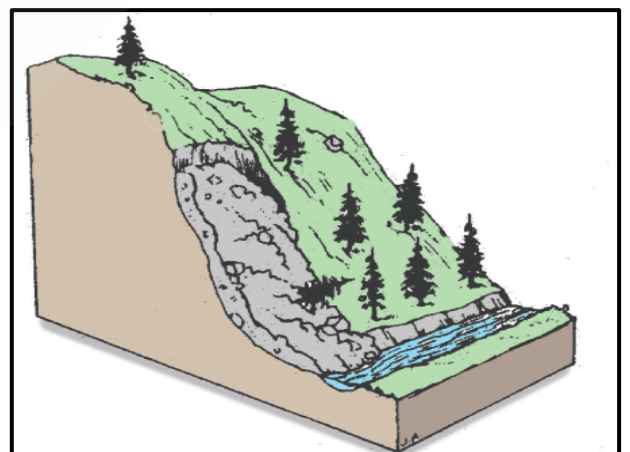
- **Overloading** - Adding weight to the top of a slope, such as through **construction or the accumulation of water**.
- **Undercutting** - Removal of support at the base of a slope, often by natural processes like **river erosion** or human activities such as **mining**.
- **Water Saturation** - Increased pore **water pressure** reduces the friction between particles, making the slope more susceptible to failure.
- **Earthquakes** - Seismic activity can induce slumping by **shaking the ground** and weakening the slope structure.

Debris Slide:

A debris slide involves the **rapid downslope movement (without backward rotation) of unconsolidated soil, rock, and organic material**. Unlike slumps, debris slides **do not exhibit rotational movement** but instead slide along a more or less planar surface.

Key characteristics and triggers include:

- **Loose Material** - Debris slides typically involve a mixture of **soil, rock, and vegetation**.
- **Steep Slopes** - Occur on steep inclines where **gravitational forces are significant**.
- **Rainfall** - Heavy or prolonged rainfall can saturate the material, **reducing cohesion** and increasing the likelihood of sliding.
- **Disturbance** - Natural events like **earthquakes or human activities** such as deforestation and construction can destabilize the slope.

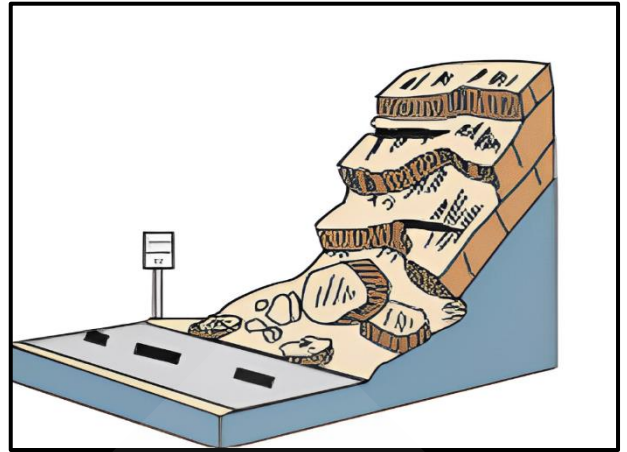


Debris Fall:

A debris fall is a **sudden, free-fall movement** of rock, soil, and debris from a steep slope or cliff.

This type of landslide is characterized by:

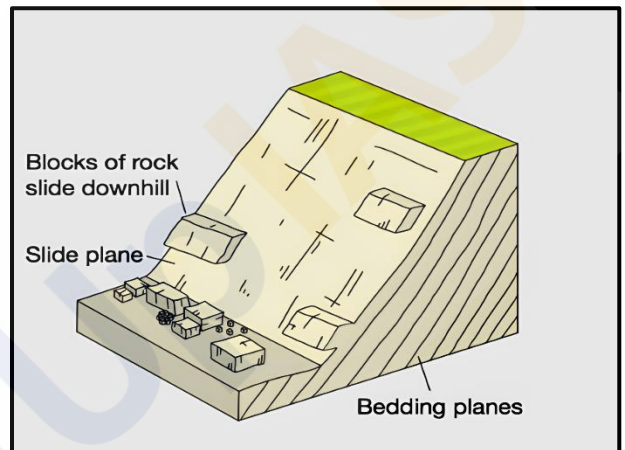
- **Free-Fall Motion** - Material falls freely through the air and accumulates at the base of the slope.
- **Steep Terrain** - Typically occurs on **vertical or near-vertical slopes**.
- **Triggering Events** - Earthquakes, weathering, and freeze-thaw cycles can dislodge material.
- **Accumulation Zone** - The fallen debris forms a talus or scree slope at the base of the cliff.

**Rockslide:**

A rockslide involves **the rapid downslope movement of large blocks of bedrock**.

This type of landslide is distinguished by:

- **Block Movement** - Large, coherent blocks of rock slide along a defined planar surface or series of surfaces.
- **Structural Weaknesses** - Joints, faults, and bedding planes within the rock provide pathways for sliding.
- **Triggers** - Earthquakes, weathering, and the removal of support at the base of the slope can initiate rockslides.
- **High Velocity** - Rockslides can move at **high speeds** and travel long distances, posing significant hazards.

**Factors Affecting Landslides:****Geological Factors:**

- **Rock Type** - Soft or fractured rocks are more prone to landslides.
- **Structural Features** - Faults, joints, and bedding planes can act as slip surfaces.

Topographical Factors:

- **Slope Angle** - Steeper slopes are more susceptible to landslides.
- **Elevation and Relief** - Higher elevations and significant relief differences can increase the potential for landslides.

Hydrological Factors:

- **Rainfall** - Intense or prolonged rainfall can saturate the ground and trigger landslides.
- **Groundwater** - High groundwater levels reduce slope stability.

Ground water near to water surface more chance of solifaction.

Ex. Bihar, 1934

Vegetation:

- **Root Systems** - Vegetation can stabilize slopes through root reinforcement.
- **Deforestation** - Removal of vegetation reduces slope stability and increases landslide risk.

Human Activities:

- **Construction** - Excavation and construction can destabilize slopes.
- **Mining and Quarrying** - Extractive activities can remove support from slopes.
- **Land Use Changes** - Changes in land use, such as urbanization, can increase the risk of landslides.

Landslide Prevention and Mitigation:

Slope Stabilization:

- **Retaining Walls** - Constructing retaining walls to support and stabilize slopes.
- **Terracing** - Creating terraces to reduce slope angles and minimize erosion.

Drainage Control:

- **Surface Drains** - Installing surface drains to divert water away from slopes.
- **Subsurface Drains** - Installing subsurface drains to reduce groundwater pressure.

Reforestation:

- **Planting Vegetation** - Establishing deep-rooted plants to reinforce the soil.
- **Bioengineering** - Using plants and natural materials to stabilize slopes.

Land Use Planning:

- **Zoning Regulations** - Implementing zoning regulations to restrict development in high-risk areas.
- **Risk Mapping** - Creating landslide hazard maps to guide land use planning and development.

Monitoring and Early Warning Systems:

- **Instrumental Monitoring** - Using sensors and instruments to monitor slope stability and detect early signs of movement.
- **Early Warning Systems** - Developing early warning systems to alert communities of impending landslides.

Engineering Solutions:

- **Rock Bolts and Anchors** - Installing rock bolts and anchors to secure unstable rock masses.
- **Grouting** - Injecting grout into fractures and voids to strengthen the slope material.

Debris Avalanche and Landslide in India:

Debris avalanches and landslides are significant natural hazards in various parts of India, particularly in the Himalayas and the Western Ghats. These events can have devastating **impacts on the environment, infrastructure, and human lives**. The causes and characteristics of these hazards differ between the two regions due to their distinct geological and climatic conditions.

In the Himalayas:

The Himalayas are one of the youngest and most dynamic mountain ranges in the world. They are characterized by high seismic activity, steep slopes, and a complex geological structure. Several factors contribute to the frequent occurrence of debris avalanches and landslides in this region:

- **Tectonic Activity:**
 - The Himalayas are **tectonically active** due to the ongoing collision between the Indian Plate and the Eurasian Plate. This tectonic activity leads to **frequent earthquakes, which can trigger landslides** and avalanches by destabilizing slopes.

- **Geological Composition:**

- The Himalayas are primarily composed of **sedimentary rocks and unconsolidated or semi-consolidated deposits**. These materials are **inherently less stable than igneous or metamorphic rocks**, making the region more susceptible to slope failures.

- **Steep Slopes:**

- The slopes in the Himalayan region are extremely steep. The combination of **steep gradients** and loose geological materials significantly increases the risk of landslides and debris avalanches.

- **Weathering and Erosion:**

- Intense weathering and erosion processes, driven by **heavy rainfall, snowmelt, and temperature variations**, further weaken the slopes, contributing to the instability.

In the Western Ghats:

The Western Ghats and the **Nilgiris**, which border Tamil Nadu, Karnataka, and Kerala, are relatively tectonically stable compared to the Himalayas. However, debris avalanches and landslides still occur in this region due to several factors:

- **Tectonic Stability and Geological Composition:**

- Unlike the Himalayas, the Western Ghats are **mostly made up of very hard, crystalline rocks**. Despite being tectonically stable, the region experiences landslides due to other geomorphological and climatic factors.

- **Steep Slopes and Escarpments:**

- Many slopes in the Western Ghats and Nilgiris are steep, with **almost vertical cliffs and escarpments**. The steep terrain enhances the gravitational force acting on the slopes, making them prone to landslides.

- **Mechanical Weathering:**

- Mechanical weathering, driven by **temperature fluctuations**, is significant in the Western Ghats. The **expansion and contraction of rocks** due to temperature changes lead to the development of cracks and fractures, weakening the rock structure over time.

- **Heavy Rainfall:**

- The Western Ghats receive **heavy rainfall, particularly during the monsoon season**. The intense and prolonged rainfall saturates the soil and rock, reducing their shear strength and leading to slope failures. The high rainfall also contributes to the formation of landslide-prone debris flows.

Erosion:

Erosion is a natural process involving the **detachment and removal of loosened rock materials and soils by external processes**, free from human interference, and is also known as **geological erosion**. The slow removal of soil is a fundamental aspect of the geological process of **denudation**, which is both inevitable and universal.

Accelerated erosion refers to the **increased rate of erosion caused by human-induced land-use changes**. Therefore, when we refer to soil erosion, we usually mean accelerated erosion, also known as **man-induced erosion**.

Soil Erosion:

Soil erosion involves the **loosening and displacement of topsoil particles**. This can occur as a slow, natural process (geological erosion) or as a rapid process driven by **deforestation, floods, tornadoes**, or other **human activities**. Soil erosion represents an **extreme form of soil degradation**, where the natural geomorphological process is accelerated to such an extent that soil is removed at a rate ten to several thousand times faster than under conditions of natural vegetation, far outpacing the rate of new soil formation.

Accelerated soil erosion, or man-induced soil erosion, is particularly **prevalent in humid climate regions where extensive forest clearance, grassland removal, and overgrazing and trampling by livestock** have occurred at alarming rates due to human activity.

Soil erosion primarily involves two processes:

1. **Loosening and detachment of soil particles** from the soil mass
2. **Removal and transport of the detached soil particles** downslope

Soil erosion and land degradation together, constitute one of the major problems that disturb the ecological balance of the world. The rapid increase in the human population has placed a great strain on the land and soil resources resulting in land degradation and soil erosion. On a worldwide basis, more than 4.85 billion acres (1.96 billion hectares) or **17% of the earth under vegetation has been degraded by humans to various extents**.

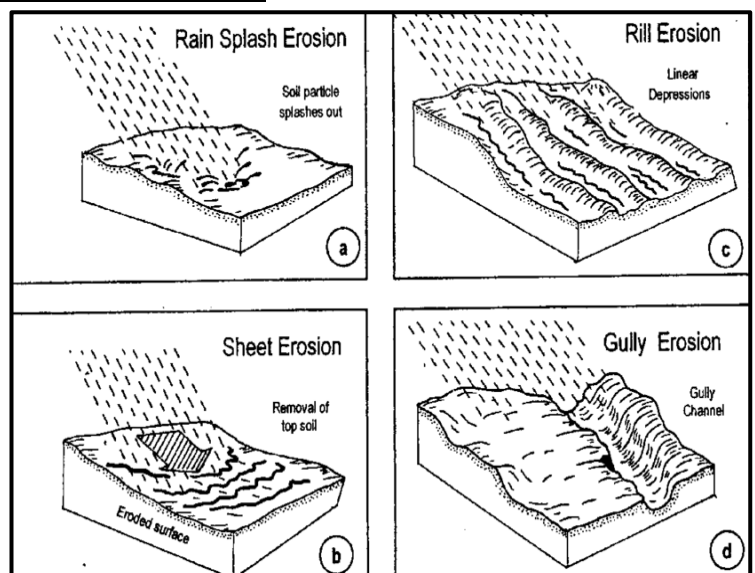
Types of Soil Erosion:

Water Erosion:

This type of erosion is caused by the **action of water**, which can remove soil through the impact of raindrops and the movement of surface water.

Depending on how the soil is lost, water erosion can be categorized as:

- **Splash Erosion** - This is the initial stage of the erosion process. It occurs when **raindrops hit bare soil with enough force to break up soil aggregates**. The explosive impact disperses individual soil particles, causing them to 'splash' onto the soil surface.
- **Sheet Erosion** - In this form, a **thin layer of soil is uniformly removed from a large area**. This type of erosion occurs when water flows over the soil surface in a dispersed manner, carrying away a consistent layer of soil.



- **Rill Erosion** - When sheet erosion progresses with greater intensity, **the runoff water flows rapidly over the soil surface, carving out well-defined, finger-shaped grooves or channels**. These small channels are known as rills and indicate a more advanced stage of erosion compared to sheet erosion.
- **Gully Erosion** - Gully erosion occurs when several rills converge, usually on a steep slope. This convergence forms wider and deeper channels called gullies, which can significantly alter the landscape.

Methods to control Water Erosion:

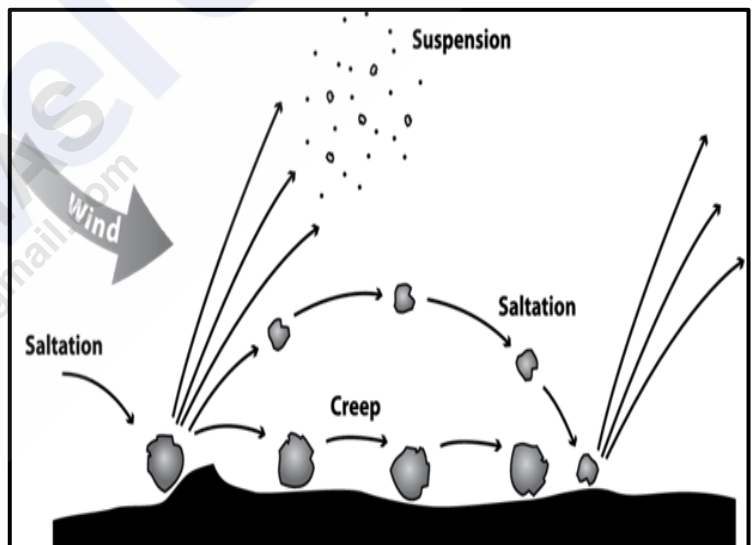
- **Contour Bunding** - Building embankments along the contours of the land to slow down water runoff and encourage water infiltration.
- **Channeling** - Constructing channels to direct water flow and reduce erosion.
- **Gully Path Modifications** - Altering the paths of gullies to control water flow and prevent further erosion.
- **Changing Cropping Patterns** - Implementing crop rotation and other agricultural practices to maintain soil cover and reduce erosion.
- **Land Leveling** - Smoothing out the land surface to reduce runoff and encourage water absorption.

Wind Erosion:

Wind is a formidable agent of erosion, driving aeolian processes that **transport dust, sand, and ash**. Wind erosion is **prevalent in arid regions where the soil is primarily sandy, and vegetation is sparse or non-existent**. When strong winds blow over exposed topsoil, it can be carried away in dust or sand storms.

There are three main types of wind erosion:

- **Saltation** - This is the primary method of soil movement by wind. In saltation, **fine soil particles are lifted into the air and drift horizontally across the surface**. These particles travel much farther horizontally than they do vertically, typically about four times as far.
- **Suspension** - Here, the smallest soil particles are carried into the air by the wind, forming a fine dust that can be transported over long distances.



These particles remain suspended in the air, moving with the wind over greater distances.

- **Surface Creep** - Heavier soil particles that are too large to be lifted into the air are pushed along the surface by the wind. This process involves the gradual movement of these larger particles across the ground.

Methods to control Wind Erosion:

- **Vegetative Cover** - Planting grasses, shrubs, or trees can help stabilize the soil and reduce wind speed near the surface. Vegetative cover **acts as a barrier to the wind** and helps bind the soil with root systems.
- **Windbreaks and Shelterbelts** - Establishing rows of trees or shrubs, known as windbreaks or shelterbelts, can reduce wind speed and protect soil. These structures can be **placed around fields or along the edges of farmland**.

- **Soil Moisture Management** - Keeping the soil moist can help reduce wind erosion since wet soil is less likely to be blown away compared to dry, loose soil. Techniques like **mulching or using irrigation systems** can help maintain soil moisture.
- **Tillage Practices** - Adopting conservation tillage practices, such as **reduced tillage or no-till farming**, helps maintain soil structure and reduces the amount of loose soil available for wind to carry away.
- **Erosion Control Mats and Mulches** - Applying materials like erosion control mats or mulches can protect soil surfaces from the wind. These materials can be made from natural or synthetic fibers and help to stabilize the soil and reduce erosion.

Glacial Erosion:

Glacial erosion refers to the process by which **glaciers shape and modify the Earth's surface through the removal and transportation of sediment and rocks**. It is influenced by various factors such as **basal sliding, tectonic rock uplift, and climate change**. In cold regions and high mountains, glaciers move slowly, transporting everything from sand to boulders.

As glaciers move, they grind against the ground, eroding both the earth and the rocks they carry. This process creates basins and **steep-sided valleys**, leaving behind visible sediment called moraine.

Ice Age glaciers shaped many landscapes, including the **Finger Lakes in New York and the fjords of Scandinavia**. Today, glaciers in **Greenland and Antarctica** continue to erode the earth, although measuring this erosion is challenging due to the ice sheets' thickness. These ice sheets erode rapidly, up to half a centimeter (0.2 inches) annually.

Factors affecting Soil Erosion:

- **Climate** - Weather conditions, including rainfall intensity and frequency, play a crucial role in erosion. Heavy rainfall can lead to increased surface runoff, which can cause soil detachment and transport. Conversely, dry climates may experience less erosion but can still be affected by wind erosion.
- **Soil Type** - Different soil types have varying susceptibilities to erosion. Sandy soils are generally more prone to erosion due to their loose structure, while clay soils are more cohesive and less easily eroded. Soil texture, structure, and organic matter content all influence erosion rates.
- **Vegetation Cover** - Vegetation acts as a natural barrier to erosion by stabilizing soil with root systems and reducing surface runoff. Areas with dense vegetation experience less erosion compared to bare or sparsely vegetated areas.
- **Topography** - The slope and shape of the land surface affect erosion. Steeper slopes tend to have higher erosion rates due to increased runoff velocity, which can carry away more soil. The shape of the land can also influence how water and wind move across it.
- **Land Use** - Human activities such as deforestation, agriculture, and construction can significantly impact erosion. Practices like ploughing, logging, and urban development can remove vegetation, disturb soil, and increase erosion rates.
- **Water Flow** - The presence and flow of water, including rivers, streams, and rainfall runoff, are major erosion factors. The speed and volume of water flow can erode riverbanks, create gullies, and transport sediment.
- **Wind** - In arid and semi-arid regions, wind can be a significant factor in erosion. Strong winds can lift and carry away loose, dry soil particles, leading to desertification and land degradation.

- **Soil Conservation Practices** - Measures such as terracing, contour ploughing, and cover cropping can help reduce erosion by managing water flow and stabilizing soil. Effective soil conservation practices can significantly mitigate erosion.
- **Geological Factors** - The underlying geology of an area, including rock type and structure, influences erosion. Softer, more erodible rocks and sediments are more easily worn away compared to harder, more resistant materials.
- **Human Activity** - Construction, mining, and other land-disturbing activities can exacerbate erosion. Alterations to the landscape, such as removing vegetation or changing natural water flow patterns, can increase erosion rates.

Positive Impacts of Soil Erosion:

- **Soil Formation** - Erosion helps in the formation of new soil layers. As materials are broken down and transported, they contribute to the development of fertile soil in other locations.
- **Nutrient Redistribution** - Erosion can move nutrient-rich sediments from one area to another, potentially enhancing soil fertility in regions where nutrients are needed.
- **Creation of Landforms** - Erosion shapes landscapes, creating various landforms such as valleys, canyons, and coastal cliffs, which contribute to natural beauty and ecological diversity.
- **Sediment Deposition** - Eroded materials often settle in new areas, forming river deltas and floodplains that can be rich in nutrients and support agriculture.
- **Aquifer Recharge** - In some cases, eroded materials can help in the recharge of aquifers by increasing the porosity of soil and allowing water to infiltrate deeper into the ground.
- **Habitat Creation** - Erosion can create diverse habitats for plants and animals, such as rocky outcrops and sediment-rich wetlands.
- **Improved Water Flow** - Erosion can sometimes lead to the formation of channels and pathways that help in managing and directing water flow, reducing flood risks.
- **Increased Biodiversity** - The changing landscape due to erosion can create niches for different species, enhancing local biodiversity.
- **Sediment Supply to Ecosystems** - Erosion provides a continuous supply of sediments to coastal and riverine ecosystems, which can be crucial for maintaining certain ecological functions.
- **Cultural and Scientific Value** - Erosion exposes geological layers and fossils, offering valuable insights into Earth's history and contributing to scientific research.

Negative Impacts of Soil Erosion:

- **Soil Loss** - Erosion leads to the loss of fertile topsoil, which can severely impact agricultural productivity and food security.
- **Decreased Land Productivity** - As soil is eroded, the remaining soil may become less productive, leading to lower crop yields and degraded farmland.
- **Water Quality Degradation** - Eroded sediments can carry pollutants into water bodies, causing water quality issues and harming aquatic life.
- **Infrastructure Damage** - Erosion can undermine roads, buildings, and other infrastructure, leading to costly repairs and potential safety hazards.

- **Increased Flood Risk** - The loss of vegetation and soil structure due to erosion can increase the risk of flooding by reducing the land's ability to absorb and retain water.
- **Loss of Agricultural Land** - Persistent erosion can lead to the loss of large areas of agricultural land, impacting local economies and food production.
- **Ecosystem Disruption** - Erosion can disrupt ecosystems by altering habitats and leading to the loss of plant and animal species that depend on stable environments.
- **Sedimentation in Waterways** - Excessive sedimentation in rivers and streams can affect water flow, reduce aquatic habitats, and increase the risk of reservoir siltation.
- **Increased Maintenance Costs** - Areas affected by erosion often require increased maintenance and conservation efforts to prevent further degradation and mitigate its impacts.
- **Economic Losses** - The overall economic impact of erosion, including reduced land value, increased maintenance costs, and loss of productivity, can be significant for communities and economies.

Methods of Controlling Soil Erosion:

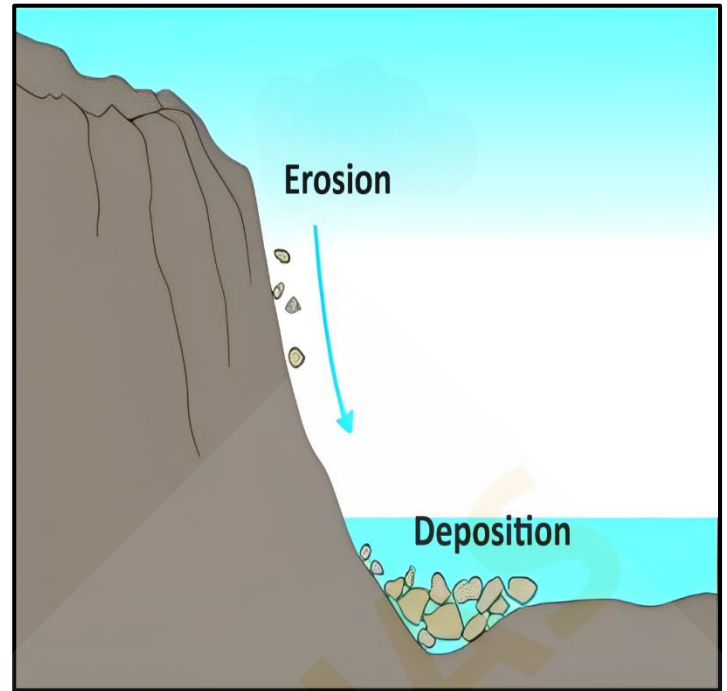
- **Vegetative Cover** - Planting grass, shrubs, or trees helps stabilize the soil with their root systems. Vegetation acts as a protective layer that reduces the impact of rainfall and slows down surface water flow, preventing soil particles from being washed away.
- **Contour Plowing** - This agricultural practice involves plowing along the contour lines of a slope rather than up and down. It helps to create natural barriers that slow down water runoff and encourage water to infiltrate into the soil, reducing erosion.
- **Terracing** - Building terraces on steep slopes creates flat areas that slow water runoff and capture soil. This method effectively reduces the speed of water flow and provides areas for vegetation to grow, further preventing erosion.
- **Cover Crops** - Growing cover crops like clover or rye during the off-season helps protect the soil from erosion. These crops keep the soil covered, reduce runoff, and their root systems help bind the soil together.
- **Reforestation and Afforestation** - Planting trees in deforested or barren areas helps to stabilize the soil. Tree roots improve soil structure and reduce the speed of surface water runoff, while tree canopies protect the soil from direct rainfall.
- **Gully Plugging** - Filling in and stabilizing eroded gullies with materials like rocks, vegetation, or constructed barriers helps prevent further erosion. Gully plugging slows water flow and encourages sediment deposition.
- **Erosion Control Blankets** - These are mats made from natural or synthetic fibers that are placed on bare soil. They protect the soil from wind and water erosion while helping to establish vegetation.
- **Building Retaining Walls** - Constructing walls on steep slopes can help prevent soil from sliding or washing away. Retaining walls can be made from various materials like stone, concrete, or timber and are often used in urban and agricultural settings.
- **Proper Drainage Systems** - Installing and maintaining effective drainage systems helps manage water flow and prevent excessive runoff. Techniques such as installing drainage ditches, French drains, or rain gardens can help direct and manage water to prevent erosion.
- **Reduced Tillage** - Minimizing tillage or practicing no-till farming helps to maintain soil structure and organic matter. Reduced tillage leaves crop residues on the soil surface, which protects the soil from erosion and helps retain moisture.

Deposition:

Deposition, a **direct consequence of erosion**, plays a vital role in shaping the Earth's surface. As erosional agents such as running water, glaciers, wind, waves, and groundwater move across the landscape, they eventually lose their velocity and energy, especially when encountering gentler slopes. This reduction in energy causes the materials they carry to settle.

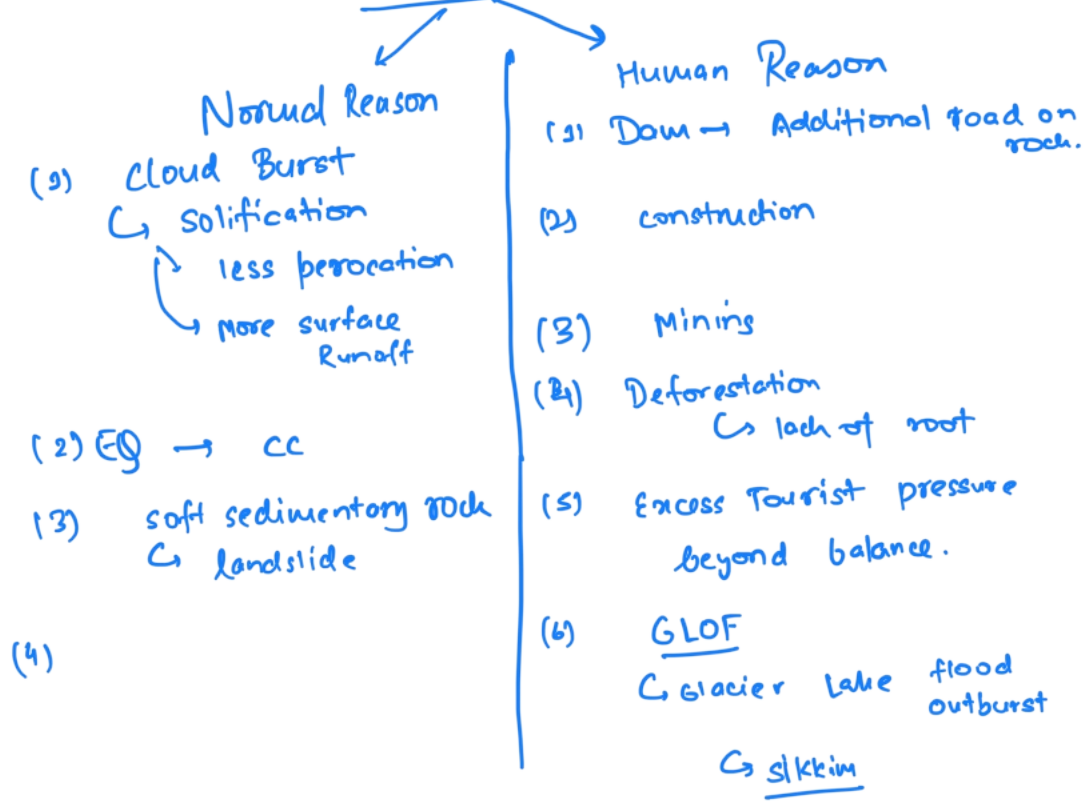
Deposition is a **passive process, occurring naturally as these transported materials are laid down**. The sequence of deposition typically begins with **coarser materials settling first, followed by finer ones**. This process often fills depressions created by previous erosional activities, contributing to the gradual levelling of the terrain.

Thus, the **same forces responsible for erosion also act as agents of deposition**, continually modifying and reshaping the Earth's surface.



===== XXXXX =====

Land slide



⇒